

INSTITUTIONAL ADOPTION · APPLIED RESEARCH

How a Research Institution Adopts Agentic AI

Design Decisions Behind Institution-Wide Use of Supervised Multi-Agent Systems at TII

Research Proposal · UAE AI Award 2026 · Category 5: AI Scientific Research

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Abstract

Agentic research often asks what an agent can do on a benchmark. Institutions face a different question: what does it take for hundreds of scientists across disciplines to adopt autonomous agents in their daily work and continue using them? This research addresses that question through an institutional deployment.

InnoviumAI is a multi-agent orchestration platform developed at the Technology Innovation Institute (TII) in Abu Dhabi. It gives researchers a shared environment for supervised agent teams connected to their scientific tools. We study how adoption spreads, whether use is sustained, and how four design decisions address barriers to institutional deployment. A desktop application bundles its runtime to simplify installation. Centralised provider access removes the need for researchers to manage shared model credentials. Human approval and recorded actions support oversight. A versioned marketplace lets teams share and adapt agents, tools and workflows while preserving local customisations.

By 15 September 2026, the platform had 433 registered accounts, with 305 recording model activity. An executive report documents 18 use cases across seven research centres, spanning scientific reporting, materials research, energy modelling and engineering simulation. Centre-reported gains include gas-turbine reporting reduced from three hours to five minutes. Applications also extend to an AI-assisted heat-exchanger design that progressed through simulation, manufacturing and laboratory testing.

The proposed study combines longitudinal adoption analysis, evaluation of the four design choices, and cross-team workflow reuse. It aims to produce institutional evidence and a tested reference architecture, supporting wider adoption at TII and a practical pathway to UAE commercialisation and government replication.

1. Research goals

The overall goal is to understand how agentic AI is adopted across a research institution and which design decisions enable that adoption. This research examines what happens when hundreds of scientists across disciplines gain access to supervised autonomous agents, and which system characteristics support continued use. This leads to two questions.

RQ1. What does institution-wide adoption of agentic AI look like? When agents are introduced into research centres, how quickly does use spread, how many disciplines does it reach, is it sustained, and does it move from individual to shared use?

We will measure registered users, monthly active users, projects created, and request volume across research centres. We will track whether early adopters remain active and how agents, tools, skills and workflows circulate through the marketplace. **Hypothesis:** adoption spreads beyond the centre that developed the platform, activity persists after the initial launch, and shared agent teams account for a growing proportion of use.

RQ2. Which design decisions remove the barriers to that adoption? Four barriers can limit institutional adoption: installation, credentials, trust and isolation. We evaluate one design decision against each.

- **Installation:** a desktop application bundles its runtime, reducing the setup required for researchers. **Hypothesis:** simpler installation helps extend adoption to disciplines where researchers do not routinely manage software environments.
- **Credentials:** shared provider credentials are managed centrally, with model access and per-user cost accounting through a common gateway. **Hypothesis:** managed access enables broader participation by removing the need for individual credential setup and administration.
- **Trust:** approval policies within the native agent loop govern tool calls and delegation, while execution records make actions reviewable. **Hypothesis:** researchers are more willing to delegate work when they can understand, approve and review consequential actions.
- **Isolation:** a versioned marketplace separates shared configurations from local overrides, allowing one centre's agent teams to be adopted and adapted by another. **Hypothesis:** preserving local customisations through shared updates helps turn individual experimentation into institutional reuse.

The contribution combines institutional adoption evidence with an explanation of the design choices behind it. The practical next question is whether this approach can be repeated in UAE government entities through the commercialisation and replication pathway described in Section 3.

2. Methodology

This is a design-science study of a deployed system. The artefact is developed, used in practice, and evaluated against the research questions using evidence generated by its deployment. Longitudinal activity records will establish adoption patterns; research-centre cases and focused evaluations will help explain how the platform's design supports those patterns.

2.1 The artefact

InnoviumAI consists of a NestJS orchestration backend, React clients, a Tauri desktop application with a bundled runtime, and a dashboard for provider access, model selection, accounting and the marketplace. Native general-purpose and Principal Investigator agents share a ReAct execution loop: reason, act, observe and repeat. The Principal Investigator agent adds milestone-specific guidance. External tools and agents connect through the Model Context Protocol (MCP). Workflows combine agents, tools, conditions and parallel branches.

The platform supports the award's five agentic properties: pursuing goals over multiple steps, decomposing work into subtasks, acting through connected tools, adapting to tool results and carrying context between steps. Human approval, rejection and cancellation provide supervision throughout the workflow. These capabilities let researchers delegate multi-step work while retaining responsibility for scientific decisions and final acceptance [1, §5.6].

2.2 Adoption study (RQ1)

InnoviumAI has been in use at TII since April 2026. We will analyse aggregate dashboard records to track adoption from launch, including account activity, request volume and later activity within registration cohorts. The study will extend this analysis to verified research-centre groupings, project activity, marketplace circulation and use of configurations created by other researchers.

The latest adoption snapshot provides a starting point:

Metric	Recorded value	Period
Registered accounts	433	15 September 2026
Accounts with recorded model activity	305	Cumulative to 15 September 2026
Accounts active in the preceding 30 days	134	Rolling window at 15 September 2026
Model requests in the launch month	2,668	April 2026, partial launch month
Model requests in the latest complete month	102,482	August 2026
Model requests across April–August	228,718	April–August 2026
Tokens processed in August	Approximately 6.11 billion	August 2026
April-registered accounts active in August	22 of 42	Registration-cohort follow-up

August request volume was 38.4 times that of the partial April launch month. The study will examine both expansion and sustained participation, distinguishing registered accounts from active use and request counts from completed research tasks [2].

The adoption is further illustrated by reported **use cases across multiple research centres**, spanning autonomous robotics, advanced materials, cryptography, directed energy, propulsion and space, quantum research, and renewable and sustainable energy. The following examples illustrate the breadth of adoption and the workflow gains reported by the centres:

Research workflow	Research centre	Reported manual time	Reported AI-assisted time
Gas-turbine controller performance report	Propulsion and Space	3 hours; 2 people	5 minutes
Scanning electron microscopy image mapping and model calibration	Quantum	4–5 weeks; 2 people	3.5 hours
Hydrogen-sector coupling and resilience model	Renewable and Sustainable Energy	3 weeks; 2 people	Approximately 4 hours
Materials image enhancement and artefact removal	Advanced Materials	Approximately 3 weeks; 1 researcher	12 hours
Greenhouse airflow and thermal simulation	Advanced Materials	Approximately 3 weeks; 1 researcher	16 hours
Research-paper improvement and review	Cryptography	Approximately 4 weeks; 2–3 people	Approximately 1 week
Lumerical simulation	Quantum	5–6 hours; 2 people	Approximately 4 hours

These are selected examples from the 18 documented use cases. Timings and staffing are centre-reported; the comparisons describe task durations rather than calculated person-hour savings [3, slides 7–8].

A further example is an AI-assisted cold-plate heat exchanger that progressed from computational design and simulation to additive manufacturing and physical laboratory testing. It illustrates how adoption can extend from routine analysis to tangible research artefacts [4].

Before InnoviumAI, researchers used separate AI assistants and manually coordinated work across different scientific tools. The shared platform brings managed model access, supervised agent execution and reusable configurations into one research environment. The adoption study will examine how researchers move from individual assistance to this shared way of working.

2.3 Barrier tests (RQ2)

Each design decision will be evaluated against the barrier it addresses.

- **Installation:** compare uptake and continued activity across disciplines with varying levels of software experience. Record onboarding effort and support needs to understand how the bundled runtime affects access.
- **Credentials:** examine provider-access configuration and per-user accounting, and record access-related onboarding or workflow interruptions. This will show how centrally managed credentials support participation in practice.
- **Trust:** exercise approval, rejection and cancellation on the nominated release, and reconstruct sampled runs from execution records and model-accounting data. Review whether researchers can understand and control the actions taken on their behalf. For external agents, distinguish approval of delegation from control over the external process's internal steps.
- **Isolation:** measure the circulation of shared configurations and their use by researchers other than the original author. Use property-based tests to check whether local overrides are preserved across marketplace updates. A second team will reuse a versioned workflow, recording setup effort, support contacts, retained customisations and the first output accepted by its domain experts.

The programme's detailed pilots offer useful cases for this evaluation. Teams reported benefits in planning, simulation setup and preliminary design, while also identifying the importance of working domain tools, explicit requirements and reliable context. Some teams built domain skills and agents from scratch. These observations sharpen the adoption question: how can a shared platform help each new team build on existing expertise? [5, slides 12–18].

For each evaluation, we will record the platform, model, tool, and configuration versions; define the relevant measures before analysis; and retain unsuccessful attempts alongside successful ones. These records will connect adoption patterns to the conditions under which researchers use the platform.

2.4 Design integrity

Architecture checks and automated tests support the platform's development and release process. The research will examine how these checks maintain the intended boundaries around model access, execution recording and approval. CI/CD validation covers builds, tests, packaged applications and deployment verification, providing a tested platform on which to conduct the adoption study. Detailed release evidence is supplied separately [6].

2.5 Limits of the evidence

The deployment study is observational. It can identify patterns of adoption and associations with design choices, but cannot by itself isolate the causal effect of each choice or establish superiority over other frameworks. Centre-reported timings provide context for use; they are not equivalent to measured institution-wide labour savings. Account-based measures will be interpreted with their window definitions and account coverage, including checks for test and service accounts. The contribution is institutional adoption evidence, supported by design analysis, focused evaluations and documented conditions for reuse.

2.6 Responsible AI

Native tool and delegation policies support human approval, notification and disabling of actions. Execution records support review, while centrally managed model access reduces the need to distribute shared provider credentials. Sensitive workloads can use self-hosted models, with connected tools and associated data flows included in deployment planning. Researchers retain responsibility for scientific acceptance and any physical testing.

The study will document data access, retention, and accountable owners, and examine whether onboarding and effective use differ across relevant technical and language groups. The UAE AI Ethics self-assessment has been completed and its results will be submitted separately.

3. Expected outcomes

1. **A validated reference architecture** for supervised multi-agent systems in research institutions, documenting how installation, managed access, oversight and adaptable shared configurations work together. Its design rationale and evaluation results will help make the approach reusable by other institutions.
2. **Institutional evidence of adoption at scale**, showing how participation spreads across disciplines, how use is sustained and how individual configurations become shared research resources. The operating platform and documented case portfolio provide the basis for this study.
3. **Commercialisation in the UAE**. The programme is developing a pathway from TII research to a UAE spin-out and its first deployments. The research will inform the onboarding, support and reusable workflow packages needed for that transition.
4. **Replication across government**. A first partner pilot will apply the approach to a bounded internal research or technical-reporting process, with defined data boundaries and acceptance criteria. The marketplace model will allow the entity to distribute vetted agent teams while staff adapt them to local needs. Over five years, the intended pathway is from an initial pilot to repeat deployments supported by a maintained catalogue of tested configurations.
5. **Wider adoption across TII**, supported by reusable domain workflows, shared expertise and consistent oversight and cost accounting. Researchers, research-centre support teams and future institutional partners will benefit from clearer guidance on how to introduce and sustain agentic AI in everyday work.

4. Relation to prior work

- **ReAct** (Yao et al., 2023) established interleaved reasoning and action. InnoviumAI builds on this pattern with a shared native execution loop and integrated oversight, evaluated through use across a research institution [7].
- **Multi-agent frameworks**, including AutoGen, CrewAI and LangGraph, provide building blocks for agent orchestration. AutoGen supports combinations of models, tools and human input; LangGraph provides execution interrupts for human intervention. InnoviumAI's contribution is the institutional integration of orchestration, managed access, oversight and configuration reuse, together with evidence of how researchers adopt that environment [8–9].
- **Model Context Protocol** (Anthropic, 2024) standardises connections between AI applications and external tools and data. InnoviumAI uses these connections within supervised workflows and makes the boundary between native execution and external-agent delegation explicit [10].
- **Human-AI interaction research**, including Amershi et al.'s *Guidelines for Human-AI Interaction* (2019), informs how users understand, direct and correct AI behaviour. This study examines these concerns in the context of sustained institutional use of agent teams [11].
- **Software product-line and configuration-override approaches** inform the separation of shared base configurations from local adaptations. The proposed study examines how that separation supports distribution and reuse of agent teams across scientific disciplines.

5. PI qualifications

Dr Robert Simon Merget leads InnoviumAI's scientific and technical direction. His experience in applied cryptography and systems security includes co-authorship of the USENIX Security 2021 *Raccoon Attack* study. This background supports the project's work on secure system design and empirical evaluation [12].

Dr Victor Juan Mateu is Chief Researcher of TII's Cryptography Research Center. His cryptographic research expertise and institutional leadership complement the project's technical and scientific direction [13].

6. Data-sharing intentions

Source code, prompts and agent configurations remain proprietary TII assets and are not planned for public release. No personal or per-user data will be shared.

To support reproducibility, we intend to share approved metric definitions, evaluation protocols, aggregate outcomes and de-identified examples. Versioned configurations, dated case reports and supporting records will be retained for controlled assessor access where authorised. This approach will allow others to examine the study's methods and findings while protecting institutional intellectual property and researcher data.

References

1. UAE AI Award 2026. [Guide to UAE AI Award 2026](#), §§5.6, 6.3 and 9.4–11.
2. TII. [Dashboard evidence and metric definitions](#), 15 September 2026. Supporting data: [monthly activity](#) and [registration cohorts](#).
3. TII. [InnoviumAI — Executive Update](#), 24 August 2026. Slides 7–8: 18 research-centre use cases and reported timing comparisons. The source filename carries 14 August; the title slide is dated 24 August.
4. TII. Cold-plate heat-exchanger case: [design workflow](#), [manufacturing and post-processing](#), and [laboratory test setup](#). Undated internal case material.
5. TII. [InnoviumAI Programme Update 26](#), reporting period 22 August–4 September 2026. Slides 12–18: pilot experiences and lessons for adoption.
6. TII. [GitLab CI/CD evidence summary](#), 15 September 2026; [production pipeline 17067](#), 14 September 2026. GitLab access requires authorisation.
7. Yao, S., et al. (2023). [ReAct: Synergizing Reasoning and Acting in Language Models](#). *ICLR 2023*.
8. Wu, Q., et al. (2023). [AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation](#). arXiv:2308.08155.
9. LangChain. [LangGraph interrupts](#). Technical documentation, accessed 15 September 2026.
10. Anthropic (2024). [Introducing the Model Context Protocol](#), 25 November 2024.
11. Amershi, S., et al. (2019). [Guidelines for Human-AI Interaction](#). *CHI 2019*.
12. Merget, R., et al. (2021). [Raccoon Attack: Finding and Exploiting Most-Significant-Bit-Oracles in TLS-DH\(E\)](#). *USENIX Security 2021*, 213–230.
13. Technology Innovation Institute. [Dr Victor Juan Mateu](#). Researcher profile, accessed 15 September 2026.